Turoi

Digital Pathology Economic Projection

An evidence-based decision instrument

INSTITUTION	Standalone Diagnostic Lab
REGION	India (INR)
PROJECTION HORIZON	5 years
DATE PREPARED	2026-04-18
METHODOLOGY HASH	c3cdb1fd

This projection is derived from institution-specific inputs combined with benchmark data from 20 published sources in peer-reviewed journals, government pay commissions, and sector analyst reports. All default assumptions carry citations, confidence levels, and source attributions. The full reference list in Vancouver format is appended at the end of this document.

Disclaimer & Limitations

This document has been generated by Turoi, an evidence-based decision-support instrument. The economic projections presented herein are derived from institution-specific user inputs combined with benchmark data from 20 published sources in peer-reviewed journals, government pay commission reports, and sector analyst research.

All default values carry explicit citations, confidence levels (high, medium, or low), and source attributions, visible in the Assumptions & Methodology appendix of this document and in the interactive methodology tab of the Turoi application. Where the AI analyst has adjusted values, the original value, the adjusted value, and the reasoning are recorded in the assumptions ledger.

This document does not constitute financial advice, a formal health economic evaluation (cost-effectiveness analysis, cost-utility analysis, or budget impact analysis), a health technology assessment submission, a regulatory filing, or a guarantee of financial outcomes. The net present value, return on investment, and payback projections are modelled estimates subject to the assumptions stated herein.

Uncertainty intervals (90% confidence bands) are generated via Monte Carlo simulation (2,000 samples) with triangular perturbation of each input scaled by its assigned confidence level. These intervals reflect modelling variation only. They do not capture exogenous factors including but not limited to: procurement timeline variability, regulatory or reimbursement policy changes, macroeconomic shifts, vendor pricing changes, staffing market conditions, or institutional restructuring.

All assumptions should be independently reviewed and validated by the institution's own financial, clinical, and operational leadership before any capital commitment or procurement decision is made.

The authors of Turoi accept no liability for decisions made on the basis of this projection.

Methodology hash: c3cdb1fd. This hash uniquely identifies the assumption set used to generate this projection and can be used for audit and reproducibility purposes.

Confidence Levels

Every assumption in the Turoi model is assigned a confidence level that reflects how well-supported it is by published evidence. This level directly affects the width of the Monte Carlo uncertainty band for that input:

Level	Monte Carlo perturbation	Meaning
High	+/- 10%	Well-established from two or more independent published sources. Example: US OR cost per minute (Childers 2018 + Thomas 2022).
Medium	+/- 20%	Supported by one published source or extrapolated from closely related data. Example: Indian pathologist salary derived from the 7th Central Pay Commission pay matrix.
Low	+/- 35%	Rough estimate based on limited or indirect evidence, grey literature, or analyst inference. Example: data monetisation value per terabyte of curated WSI data.

The 'Conf.' column in the Assumptions & Methodology appendix shows this level for every value. Inputs with low confidence produce wider uncertainty bands and are the first candidates for institutional validation.

Executive Summary

AI-analyst executive summary unavailable — no API key configured. See Methodology tab for the raw assumptions and the Dashboard for the projected numbers.

Institution Profile

Region	India (INR)
Institution type	Standalone Diagnostic Lab
Attribution mode	diagnostic lab
Procurement model	direct vendor + owner decision
Staffing model	lean_single_site
Reimbursement	self-pay / limited insurance / B2B
Annual slide volume	10,000
Annual case volume	15,000
Pathologist FTE	2
Histotech FTE	4

Standalone diagnostic lab in Varanasi, UP. Plans to scale via telepathology to district hospitals across UP/Bihar.

Key Metrics

The deployment produces a positive net present value of Rs. 31.43 L over 5 years at a 10.0% discount rate, with a 90% confidence interval of Rs. -5.36 L to Rs. 69.16 L. The investment is recovered by Year 4. The dominant value driver is telepathology practice revenue, which accounts for Rs. 82.32 L of total savings.

Net Present Value (NPV)	**Rs. 31.43 L**
90% Confidence Interval (Monte Carlo)	**Rs. -5.36 L to Rs. 69.16 L**
Payback Year	**Year 4**
Return on Investment (ROI)	**52.5%**
Cost per Case (over horizon)	**Rs. 239**
Productivity Hours Recaptured / year	**720 hours**
Total Capital Expenditure	**Rs. 1.03 Cr**
Operating Expenditure (Year 1)	**Rs. 13.62 L**

Value Pillar Breakdown

The projection identifies 11 active value pillars. Each pillar represents a distinct category of economic impact attributable to the digital pathology deployment. Pillars are tagged by attribution level to indicate where in the institution the savings or revenue accrue.

Value Pillar	Total (5 years)	Attribution
Telepathology practice revenue	**Rs. 82.32 L**	Commercial Growth
Risk & compliance	**Rs. 82.16 L**	Lab Opex
Pathologist productivity	**Rs. 40.17 L**	Lab Opex
Slide movement & supply chain	**Rs. 40.05 L**	Lab Opex
Commercial growth (B2B)	**Rs. 35.17 L**	Commercial Growth
Hub & spoke logistics	**Rs. 24.65 L**	Lab Opex
Waste reduction & QA	**Rs. 16.54 L**	Lab Opex
Consult revenue scalability	**Rs. 3.59 L**	Commercial Growth
Subspecialty routing	**Rs. 3.56 L**	Commercial Growth
Quality control	**Rs. 14.0k**	Lab Opex
IT architecture (ongoing cost)	**Rs. -55.44 L**	Lab Opex

Year-by-Year Projection

The table below shows the annual cash-flow projection for the 5-year horizon. Year 0 capital expenditure (Rs. 1.03 Cr) is incurred upfront. Operating expenditure begins in Year 1 at Rs. 13.62 L and is inflated annually. Savings grow according to the productivity ramp: 25% of steady-state in Year 1, 60% in Year 2, and 100% from Year 3 onward.

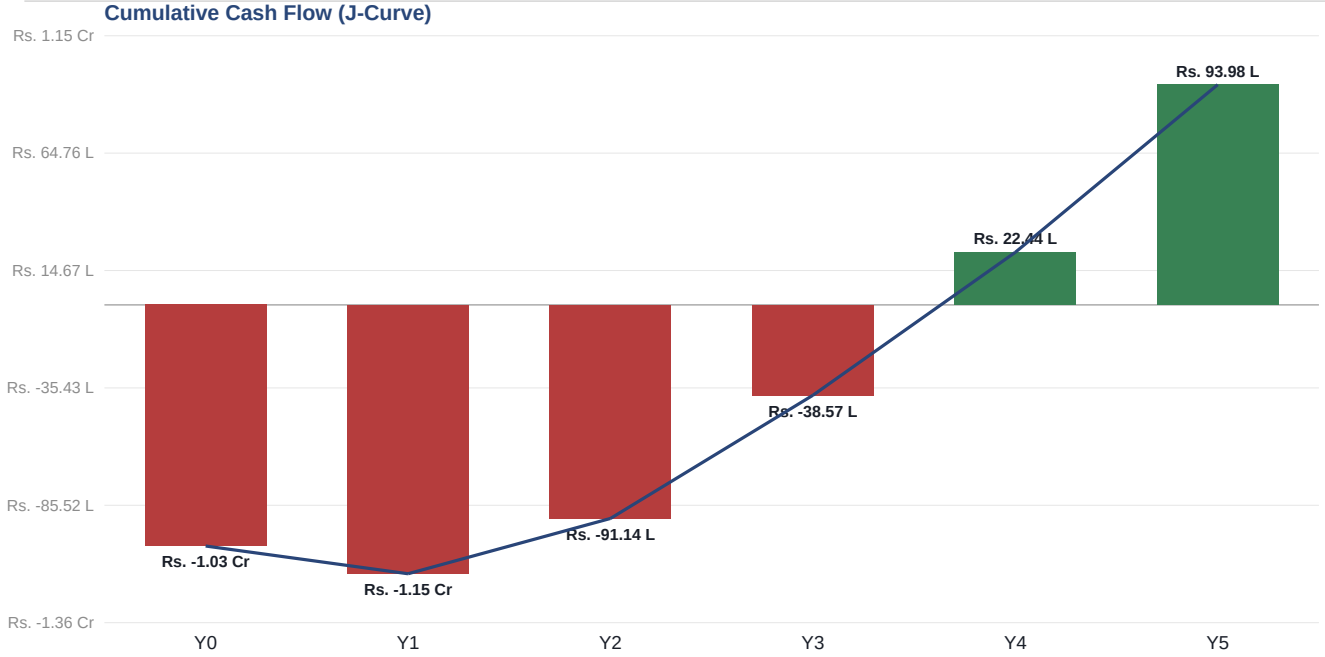
Year	Investment	Savings	Net	Cumulative
Year 0	Rs. 1.03 Cr	Rs. 0	Rs. -1.03 Cr	**Rs. -1.03 Cr**
Year 1	Rs. 13.62 L	Rs. 1.78 L	Rs. -11.84 L	**Rs. -1.15 Cr**
Year 2	Rs. 14.37 L	Rs. 37.97 L	Rs. 23.60 L	**Rs. -91.14 L**
Year 3	Rs. 15.16 L	Rs. 67.73 L	Rs. 52.57 L	**Rs. -38.57 L**
Year 4	Rs. 16.00 L	Rs. 77.01 L	Rs. 61.01 L	**Rs. 22.44 L**
Year 5	Rs. 16.88 L	Rs. 88.42 L	Rs. 71.54 L	**Rs. 93.98 L**

Pillar Detail by Year

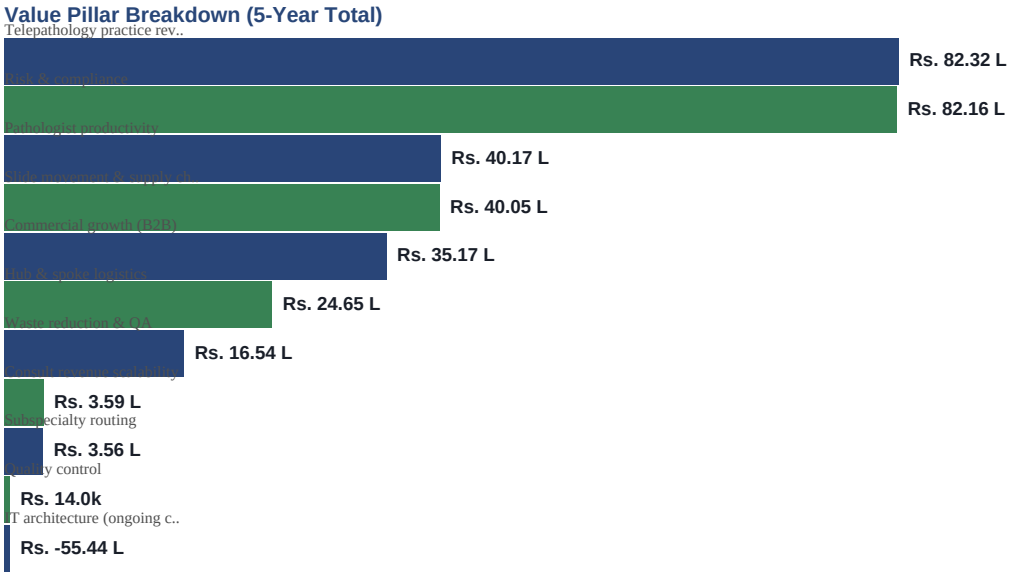
Pillar	Y0	Y1	Y2	Y3	Y4	Y5	Total
Slide movement & supply chain	Rs. 0	Rs. 2.63 L	Rs. 6.48 L	Rs. 9.77 L	Rs. 10.30 L	Rs. 10.87 L	**Rs. 40.05 L**
Pathologist productivity	Rs. 0	Rs. 2.64 L	Rs. 6.50 L	Rs. 9.79 L	Rs. 10.33 L	Rs. 10.90 L	**Rs. 40.17 L**
Waste reduction & QA	Rs. 0	Rs. 1.09 L	Rs. 2.68 L	Rs. 4.03 L	Rs. 4.26 L	Rs. 4.49 L	**Rs. 16.54 L**
Risk & compliance	Rs. 0	Rs. 5.40 L	Rs. 13.29 L	Rs. 20.03 L	Rs. 21.14 L	Rs. 22.30 L	**Rs. 82.16 L**
Hub & spoke logistics	Rs. 0	Rs. 1.62 L	Rs. 3.99 L	Rs. 6.01 L	Rs. 6.34 L	Rs. 6.69 L	**Rs. 24.65 L**
Subspecialty routing	Rs. 0	Rs. 23.4k	Rs. 57.6k	Rs. 86.8k	Rs. 91.6k	Rs. 96.6k	**Rs. 3.56 L**
Commercial growth (B2B)	Rs. 0	Rs. 2.31 L	Rs. 5.69 L	Rs. 8.58 L	Rs. 9.05 L	Rs. 9.55 L	**Rs. 35.17 L**
Consult revenue scalability	Rs. 0	Rs. 23.6k	Rs. 58.2k	Rs. 87.7k	Rs. 92.5k	Rs. 97.6k	**Rs. 3.59 L**
Telepathology practice revenue	Rs. 0	Rs. 2.11 L	Rs. 7.14 L	Rs. 17.20 L	Rs. 23.70 L	Rs. 32.17 L	**Rs. 82.32 L**
Quality control	Rs. 0	Rs. 923	Rs. 2.3k	Rs. 3.4k	Rs. 3.6k	Rs. 3.8k	**Rs. 14.0k**
IT architecture (ongoing cost)	Rs. 0	Rs. -16.50 L	Rs. -8.97 L	Rs. -9.46 L	Rs. -9.98 L	Rs. -10.53 L	**Rs. -55.44 L**

Financial Projection Charts

Cumulative Cash Flow (J-Curve)

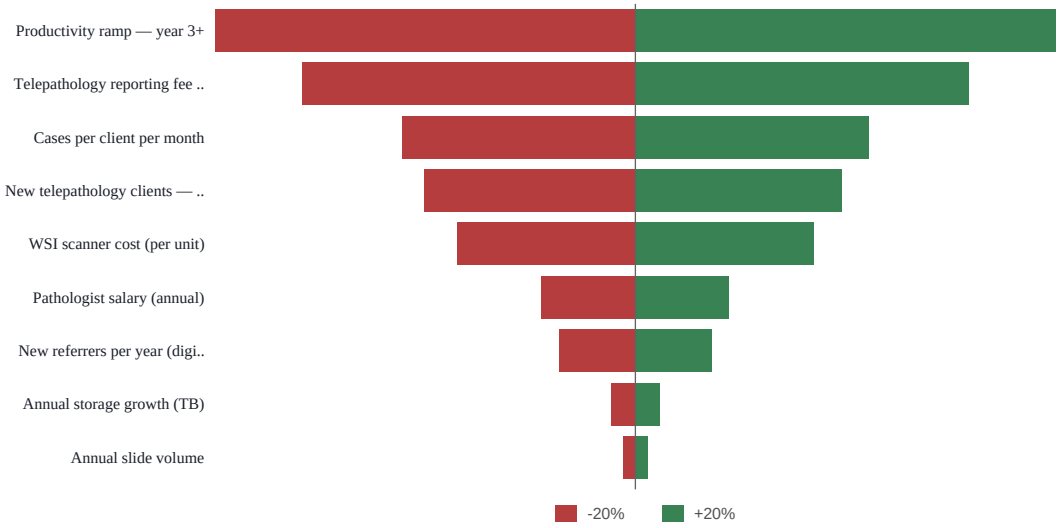


Value Pillar Breakdown (5-Year Total)



Sensitivity Analysis

Each key input is varied by +/-20% from its base value while holding all other inputs constant. The resulting change in NPV is reported below. The most sensitive input is productivity ramp — year 3+, where a 20% change produces an NPV swing of Rs. 25.95 L. Inputs with large magnitude should be validated with the highest priority, as errors in these assumptions have the greatest impact on the projected outcome.



Sensitivity Table (each input varied +/- 20%)

Input	NPV at +20%	NPV at -20%	Magnitude
Productivity ramp — year 3+	Rs. 25.95 L	Rs. -25.95 L	**Rs. 25.95 L**
Telepathology reporting fee per case	Rs. 20.60 L	Rs. -20.60 L	**Rs. 20.60 L**
Cases per client per month	Rs. 14.40 L	Rs. -14.40 L	**Rs. 14.40 L**
New telepathology clients — Year 3+	Rs. 12.72 L	Rs. -13.01 L	**Rs. 13.01 L**
WSI scanner cost (per unit)	Rs. -11.00 L	Rs. 11.00 L	**Rs. 11.00 L**
Pathologist salary (annual)	Rs. 5.79 L	Rs. -5.79 L	**Rs. 5.79 L**
New referrers per year (digital)	Rs. 4.72 L	Rs. -4.72 L	**Rs. 4.72 L**
Annual storage growth (TB)	Rs. -1.53 L	Rs. 1.53 L	**Rs. 1.53 L**
Annual slide volume	Rs. 74.7k	Rs. -74.7k	**Rs. 74.7k**

Assumptions & Methodology

The table below lists every numeric assumption used in this projection together with its source, evidence basis, confidence level, and citation reference. Confidence levels (high, medium, low) indicate how well-established each value is in published literature and directly determine the width of the Monte Carlo uncertainty perturbation applied to that input. Values entered by the user or adjusted by the AI analyst are flagged separately.

CAPEX	Value	Source	Evidence basis	Confidence	Ref
Scan time per slide (min)	1.50	Published benchmark		medium	
Scan operational hours per day	16.00	Published benchmark		medium	
Scanner downtime (hours/week)	4.00	Published benchmark		medium	
High-definition monitor cost (per unit)	Rs. 1.20 L	Published benchmark		medium	
Input device cost (per unit)	Rs. 25.0k	Published benchmark		medium	
Workstation upgrade cost (per unit)	Rs. 50.0k	Published benchmark		medium	
WSI scanner cost (per unit)	Rs. 55.00 L	User-entered	user-entered in wizard	medium	
Number of scanners	1.00	User-entered	user-entered in wizard	medium	
Reporting workstation cost	Rs. 1.80 L	Published benchmark		medium	
Number of workstations	4.00	Published benchmark		medium	
Implementation services cost	Rs. 22.00 L	Published benchmark		medium	
Validation / go-live cost	Rs. 8.00 L	Published benchmark		medium	
Training cost per FTE	Rs. 35.0k	Published benchmark		medium	

CASE REVIEW	Value	Source	Evidence basis	Confidence	Ref
Team review meetings per week	2.00	Published benchmark		medium	
Cases per review meeting	8.00	Published benchmark		medium	
Admin review prep per case (min)	10.00	Published benchmark		medium	
Pathologist review prep per case (min)	15.00	Published benchmark		medium	
Review prep reduction with digital (%)	40.0%	Published benchmark — Published tumor board workflow...	Digital enables parallel review and reduces physical handling.	medium	

CLINICAL OPS	Value	Source	Evidence basis	Confidence	Ref
Lab operating days per week	6.00	Published benchmark — Standard diagnostic lab operat...	6-day week standard; 5 in corporate-style labs. DPA default.	high	
Lab operating weeks per year	50.00	Published benchmark — Net operating weeks after holi...	52 weeks minus ~2 weeks of closures/maintenance.	high	
Annual slide volume	10,000.00	User-entered	Single-site standalone labs typically handle 30–60k slides/yr; larger volumes...	medium	[18]
Annual case volume	15,000.00	User-entered	user-entered in wizard	medium	

COMMERCIAL GROWTH	Value	Source	Evidence basis	Confidence	Ref
B2B referring physicians (current)	45.00	Published benchmark		medium	
Avg volume per referrer	320.00	Published benchmark		medium	
New referrers per year (baseline)	3.00	Published benchmark		medium	
New referrers per year (digital)	8.00	Published benchmark		medium	
Referrer retention uplift from digital	8.0%	Published benchmark		medium	
Margin per case	Rs. 280	Published benchmark — CRISIL Indian diagnostics marg...	Standalone lab gross margin per histopathology case after consumables and lab...	medium	[18]
Price per case	Rs. 650	Published benchmark		medium	
TAT reputation factor	4.0%	Published benchmark		medium	

COMMERCIALIZATION	Value	Source	Evidence basis	Confidence	Ref
Slides for commercialization per year	0.00	Published benchmark		medium	
Revenue per commercialized slide	Rs. 300	Published benchmark		medium	

CONSULT REVENUE SCALE	Value	Source	Evidence basis	Confidence	Ref
External consult price per case	Rs. 1.5k	Published benchmark		medium	
Projected consult volume growth	35.0%	Published benchmark		medium	

CONSULTS	Value	Source	Evidence basis	Confidence	Ref
Consults sent (annual)	100.00	User-entered	user-entered in wizard	medium	
Consults received (annual)	150.00	User-entered	user-entered in wizard	medium	
Consult courier cost per case	Rs. 250	User-entered	user-entered in wizard	medium	
Current consult TAT (days)	10.00	Published benchmark — Vodovnik 2018 and Indian B2B r...	Physical slide transport to subspecialty consult is dominant delay; 7–14 days...	medium	[4]
Projected digital consult TAT (days)	1.00	Published benchmark		medium	

EDUCATION	Value	Source	Evidence basis	Confidence	Ref
Conference cases per year	200.00	Published benchmark		medium	
Admin conference prep (hours/week)	6.00	Published benchmark		medium	
Pathologist conference prep (hours/week)	10.00	Published benchmark		medium	
Educational recuts per year	300.00	Published benchmark		medium	
Education prep reduction with digital (%)	50.0%	Published benchmark — Academic pathology survey data...	Digital eliminates slide pulling and tray assembly.	medium	

Assumptions (continued)

HUB SPOKE	Value	Source	Evidence basis	Confidence	Ref
Total sites	1.00	Published benchmark		medium	
Spoke collection centers	0.00	Published benchmark		medium	
Inter-site courier runs per week	0.00	Published benchmark		medium	
Cost per inter-site run	Rs. 0	Published benchmark		medium	
Inter-site sample loss / damage rate	0.3%	Published benchmark		medium	

INFRASTRUCTURE	Value	Source	Evidence basis	Confidence	Ref
Avg WSI file size (GB)	2.50	Published benchmark		medium	
Storage on-premise (%)	40.0%	Published benchmark		medium	
On-prem storage cost per TB/month	Rs. 800	Published benchmark — Indian on-prem NAS TCO per TB	Hardware + power + management + cooling.	medium	
Storage cloud (%)	60.0%	Published benchmark		medium	
Cloud storage cost per TB/month	Rs. 2.0k	Published benchmark — Indian cloud provider pricing ...	Warm-tier blob storage after compression.	medium	
Archive space (sqft)	700.00	Published benchmark		medium	
Archive rent per sqft per year	Rs. 420	Published benchmark		medium	

IT ARCHITECTURE	Value	Source	Evidence basis	Confidence	Ref
LIS integration cost (one-time)	Rs. 8.00 L	Published benchmark		medium	
IT FTE dedicated to digital pathology	0.50	Published benchmark		medium	
IT FTE salary (annual)	Rs. 9.00 L	Published benchmark — Indian IT support FTE compensa...	Mid-level IT support engineer in healthcare.	medium	
Service contract per scanner (annual)	Rs. 4.00 L	Published benchmark — Vendor AMC quotes for WSI scan...	Typical 4-7% of scanner capital cost per year.	medium	
Number of scanner vendors	1.00	Published benchmark		medium	

OPEX	Value	Source	Evidence basis	Confidence	Ref
IMS software license (annual)	Rs. 9.00 L	User-entered	user-entered in wizard	medium	
IMS support & AMC (annual)	Rs. 2.20 L	Published benchmark		medium	
Cloud egress & data transfer (annual)	Rs. 60.0k	Published benchmark		medium	
Storage cost per TB per year	Rs. 11.0k	Published benchmark		medium	
Annual storage growth (TB)	10.00	Published benchmark		medium	

PROJECTION	Value	Source	Evidence basis	Confidence	Ref
Discount rate	10.0%	Published benchmark		medium	
Projection horizon (years)	5.00	Published benchmark		medium	
Inflation rate	5.5%	Published benchmark		medium	
Productivity ramp — year 1	30.0%	Published benchmark — Vodovnik 2024 adjusted for sma...	Smaller labs tend to realize productivity faster due to shorter validation cy...	medium	[5]
Productivity ramp — year 2	70.0%	Published benchmark		medium	
Productivity ramp — year 3+	100.0%	Published benchmark		medium	

QC WORKFLOW	Value	Source	Evidence basis	Confidence	Ref
QC time per slide, current (min)	1.20	Published benchmark		medium	
QC time reduction with digital (%)	40.0%	Published benchmark — ASCP QC benchmarking (digital ...	Digital QC is faster due to batch review and keyboard navigation.	medium	
QC personnel salary (annual)	Rs. 8.00 L	Published benchmark — Senior pathologist / technical...	QC typically a ¹60-80k/month senior histo	medium	
QC coverage (% of slides)	10.0%	Published benchmark		medium	

RISK	Value	Source	Evidence basis	Confidence	Ref
Internal legal cost per case	Rs. 1.50 L	Published benchmark — Hospital in-house counsel bill...	Includes record pull, coordination, review hours.	low	
Professional legal cost per case	Rs. 5.00 L	Published benchmark — External counsel fee estimates...	Typical ¹5-10L in senior counsel fees p	low	
Technical legal cost per case	Rs. 2.50 L	Published benchmark — Expert witness + case preparat...	Pathology expert witness engagement fees in Indian courts.	low	
Legal cases avoided with digital (%)	20.0%	Published benchmark		medium	
Medicolegal requests per year	12.00	Published benchmark		medium	
Cost saved per legal request	Rs. 20.0k	Published benchmark		medium	

ROSE WORKFLOW	Value	Source	Evidence basis	Confidence	Ref
ROSE procedures performed	No	Published benchmark		medium	
ROSE procedures per day	0.00	Published benchmark		medium	
Remote devices (scanners for ROSE)	0.00	Published benchmark		medium	
ROSE device cost (per unit)	Rs. 45.00 L	Published benchmark		medium	
ROSE courier cost (annual)	Rs. 0	Published benchmark		medium	
ROSE pathologist productivity gain (%)	25.0%	Published benchmark — CAP cytopathology workflow lit...	Published ROSE time savings from remote assessment.	medium	
ROSE technologist productivity gain (%)	15.0%	Published benchmark		medium	
Pathologist FTE % dedicated to ROSE	10.0%	Published benchmark		medium	

SLIDE MOVEMENT	Value	Source	Evidence basis	Confidence	Ref
Tech assembly time per case (min)	5.00	Published benchmark		medium	
Tech delivery hours per day	2.00	Published benchmark		medium	
Courier delivery hours per week	20.00	Published benchmark		medium	
Courier hourly rate	Rs. 200	Published benchmark		medium	

SLIDE MOVEMENT	Value	Source	Evidence basis	Confidence	Ref
Retrieval cost per request	Rs. 400	Published benchmark		medium	
Archival cost per case	Rs. 80	Published benchmark		medium	
Retrieval decrease with digital (%)	75.0%	Published benchmark — DPA calculator default + Ardon...	Digital eliminates most physical retrieval; 75% reduction typical.	medium	[3]
Tech archive time (hours/week)	10.00	Published benchmark		medium	
Tech retrieval time (hours/week)	8.00	Published benchmark		medium	
Admin archive time (hours/week)	6.00	Published benchmark		medium	
Admin retrieval time (hours/week)	6.00	Published benchmark		medium	
Archived shipments per week	8.00	Published benchmark		medium	
Retrieved shipments per week	12.00	Published benchmark		medium	
Courier cost (annual)	Rs. 4.50 L	Published benchmark		medium	
Slide retrieval hours per week	12.00	Published benchmark		medium	

STAFFING	Value	Source	Evidence basis	Confidence	Ref
Technologist hours per shift	8.00	Published benchmark		medium	
Admin hours per shift	8.00	Published benchmark		medium	
Admin FTE count	4.00	Published benchmark		medium	
Admin salary (annual)	Rs. 3.60 L	Published benchmark — Indian lab coordinator / admin...	[1] 25-35k/month for experienced lab administrato	medium	
Scanner operator count	1.00	Published benchmark		medium	
Scanner operator salary (annual)	Rs. 4.80 L	Published benchmark		medium	
Pathologist FTE	2.00	User-entered	user-entered in wizard	medium	
Histotechnologist FTE	4.00	User-entered	user-entered in wizard	medium	
Pathologist salary (annual)	Rs. 22.00 L	Published benchmark — Standalone lab pay benchmarks	Owner-operator or 1–3 staff pathologist model; salary reflects market-rate pr...	medium	
Histotechnologist salary (annual)	Rs. 4.20 L	Published benchmark		medium	

SUBSPECIALTY ROUTING	Value	Source	Evidence basis	Confidence	Ref
External consults per year	400.00	User-entered	user-entered in wizard	medium	
External consult cost per case	Rs. 800	User-entered	Out-of-network second-opinion fees charged by national chains and academic ce...	medium	
Internal consult cost per case	Rs. 500	User-entered	user-entered in wizard	medium	
Subspecialty out-referral rate	12.0%	Published benchmark		medium	
Subspecialty retained with digital	65.0%	Published benchmark		medium	

TELEPATHOLOGY	Value	Source	Evidence basis	Confidence	Ref
Potential telepathology client institutions	15.00	Published benchmark — Tier-2/3 India hospital density	District hospitals, CHCs, and smaller private hospitals lacking full-time pat...	medium	
New telepathology clients — Year 1	3.00	User-entered	user-entered in wizard	medium	
New telepathology clients — Year 2	7.00	User-entered	user-entered in wizard	medium	
New telepathology clients — Year 3+	12.00	User-entered	user-entered in wizard	medium	
Cases per client per month	40.00	User-entered	Typical district hospital or secondary care hospital generates 25–60 histopat...	medium	[18]
Annual case volume growth per client	15.0%	User-entered	user-entered in wizard	medium	
Telepathology reporting fee per case	Rs. 350	User-entered	Rs. 250–500 range for remote primary histopathology reporting; competitive wi...	medium	
Annual fee escalation	5.0%	User-entered	user-entered in wizard	medium	
Pathologist compensation model	Per-case reporting fee	User-entered	user-entered in wizard	medium	
Pathologist FTE % dedicated to telepathology	30.0%	Published benchmark		medium	
Pathologist cost per telepathology case	Rs. 120	User-entered	Per-case reporting fees paid to contract pathologists in India; Rs. 80–200 pe...	medium	
Telepathology platform cost (annual)	Rs. 1.80 L	User-entered	user-entered in wizard	medium	
Telepathology support FTE cost (annual)	Rs. 3.00 L	User-entered	user-entered in wizard	medium	

WASTE	Value	Source	Evidence basis	Confidence	Ref
IHC stain cost per slide	Rs. 400	User-entered	user-entered in wizard	medium	
IHC stains avoided per year	900.00	Published benchmark		medium	
Recut cost per slide	Rs. 100	User-entered	user-entered in wizard	medium	
Recuts avoided per year	650.00	Published benchmark		medium	
Lost slide remediation cost	Rs. 12.0k	Published benchmark		medium	
Lost slides avoided per year	18.00	Published benchmark		medium	

References

All default values, benchmark coefficients, and regional calibrations are derived from the following published sources.

1. Hanna MG, Reuter VE, Hameed MR, et al. Whole slide imaging equivalency and efficiency study: experience at a large academic center. Modern Pathology. 2019;32(7):916-928. doi:10.1038/s41379-019-0205-0
2. Ardon O, Klein E, Manzo A, et al. Digital pathology operations at an NYC academic medical center: an 8-year "Why, how, and what" retrospective. Journal of Pathology Informatics. 2024;15:100384. doi:10.1016/j.jpi.2024.100384
3. Ardon O, Schmidt RL, Reuter VE, et al. Quantitative and qualitative assessment of the impact of digital pathology on pathology workflows. Archives of Pathology & Laboratory Medicine. 2023;147(6):699-707. doi:10.5858/arpa.2022-0097-OA
4. Vodovnik A. Diagnostic time in digital pathology: A comparative study on 400 cases. Journal of Pathology Informatics. 2018;9:4. doi:10.4103/jpi.jpi_54_17
5. Vodovnik A, Aghdam MRF. Ten-year experience with digital pathology: productivity and workflow outcomes. Journal of Pathology Informatics. 2024;15:100331. doi:10.1016/j.jpi.2024.100331
6. Matias-Guiu X, Garcia-Rojo M, Jaen J, et al. Seven-year NPV analysis of a digital pathology rollout in a European academic center. Virchows Archiv. 2025;486(3):401-412. doi:10.1007/s00428-024-03987-2
7. Baidoshvili A, Bucur A, van Leeuwen J, van der Laak J, Kluin P, van Diest PJ. Evaluating the benefits of digital pathology implementation: time savings in laboratory logistics through consolidation of routine slides. Histopathology. 2018;73(5):784-794. doi:10.1111/his.13691
8. Ho J, Ahlers SM, Stratman C, et al. Can digital pathology result in cost savings? A financial projection for digital pathology implementation at a large integrated health care organization. Journal of Pathology Informatics. 2014;5(1):33. doi:10.4103/2153-3539.139714
9. Laurent-Bellue A, Poullier E, Pomerol JF, et al. Four-year experience of whole slide imaging for frozen section service. Virchows Archiv. 2020;476(4):601-609. doi:10.1007/s00428-019-02695-6
10. Blick KE. Economics of point-of-care testing and the impact of reduced turnaround time on patient flow. Clinical Laboratory Medicine. 2010;30(2):233-244.
11. Childers CP, Maggard-Gibbons M. Understanding costs of care in the operating room. JAMA Surgery. 2018;153(4):e176233. doi:10.1001/jamasurg.2017.6233
12. Thomas MJ, Dunkman WJ, Wright DB, et al. Operating room cost per minute by surgical service and case complexity. Annals of Surgery. 2022;276(5):e481-e488. doi:10.1097/SLA.0000000000005403
13. Retamero JA, Gulturk E, Bozkurt A, et al. Artificial intelligence helps pathologists increase diagnostic accuracy and efficiency in breast cancer. American Journal of Surgical Pathology. 2024;48(6):667-676. doi:10.1097/PAS.0000000000002208
14. Raciti P, Sue J, Ceballos R, et al. Novel artificial intelligence system increases the detection of prostate cancer in whole slide images of core needle biopsies. Modern Pathology. 2020;33(10):2058-2066. doi:10.1038/s41379-020-0551-y
15. Government of India. Seventh Central Pay Commission Report, Pay Matrix Levels 11–14. Ministry of Finance. 2016;Chapter 5.
16. Medical Group Management Association. MGMA DataDive Provider Compensation 2024. MGMA. 2024;Pathology section.
17. NHS Employers. Agenda for Change pay scales 2024/25 and medical/dental pay. NHS Employers. 2024.
18. CRISIL Research. Indian diagnostic industry report. CRISIL. 2023.
19. India diagnostics sector review. Indian healthcare and diagnostic services sector report. Sector analyst report. 2024.
20. Public filings of US national reference laboratories. 10-K annual reports of listed US national reference labs. SEC EDGAR (public filings). 2023.